

Q1 Solve for x , each of the following equations:

$$(i) \begin{vmatrix} 1 & 2+x & 3 \\ 2 & 1 & 3+x \\ 3 & 2+x & 1 \end{vmatrix} = 0$$

Sol. Given

$$\begin{vmatrix} 1 & 2+x & 3 \\ 2 & 1 & 3+x \\ 3 & 2+x & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 2+x & 3 \\ 0 & -3-2x & -3+x \\ 0 & -4-2x & -8 \end{vmatrix} = 0$$

By $R_2 - 2R_1$
 $R_3 - 3R_1$

Expanding from C_1

$$\begin{vmatrix} -3-2x & -3+x \\ -4-2x & -8 \end{vmatrix} = 0$$

$$-8(-3-2x) - (-3+x)(-4-2x) = 0$$

$$24 + 16x - (12 + 6x - 4x - 2x^2) = 0$$

$$24 + 16x - 12 - 2x + 2x^2 = 0$$

$$2x^2 + 14x + 12 = 0$$

$$x^2 + 7x + 6 = 0$$

$$x^2 + 6x + x + 6 = 0$$

$$x(x+6) + 1(x+6) = 0$$

$$(x+6)(x+1) = 0$$

$$\Rightarrow \boxed{x = -6, -1}$$

(iii)

$$\begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 9 & 9-x^2 \end{vmatrix} = 0$$

Soln.

Given

$$\begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 9 & 9-x^2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1-x^2 & 0 & 0 \\ 2 & 1 & -3 & -1 \\ 2 & 1 & 5 & 3-x^2 \end{vmatrix} = 0$$

By $C_2 - C_1$
 $C_3 - 2C_1$
 $C_4 - 3C_1$

Expanding from R_1

$$\begin{vmatrix} 1-x^2 & 0 & 0 \\ 1 & -3 & -1 \\ 1 & 5 & 3-x^2 \end{vmatrix} = 0$$

Expanding from R_1

$$(1-x^2) \begin{vmatrix} -3 & -1 \\ 5 & 3-x^2 \end{vmatrix} = 0$$

$$(1-x^2)(-9+3x^2+5) = 0$$

$$(1-x^2)(3x^2-4) = 0$$

$$1-x^2 = 0, \quad 3x^2-4 = 0$$

$$x^2 = 1, \quad x^2 = 4/3$$

$$x = \pm 1, \pm \frac{2}{\sqrt{3}}$$

$$(iii) \begin{vmatrix} 1 & x & x^2 & x^3 \\ 1 & 2 & 2^2 & 2^3 \\ 1 & 3 & 3^2 & 3^3 \\ 1 & 4 & 4^2 & 4^3 \end{vmatrix} = 0$$

Sol. Given

$$\begin{vmatrix} 1 & x & x^2 & x^3 \\ 1 & 2 & 2^2 & 2^3 \\ 1 & 3 & 3^2 & 3^3 \\ 1 & 4 & 4^2 & 4^3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x & x^2 & x^3 \\ 0 & 2-x & 2^2-x^2 & 2^3-x^3 \\ 0 & 3-x & 3^2-x^2 & 3^3-x^3 \\ 0 & 4-x & 4^2-x^2 & 4^3-x^3 \end{vmatrix} = 0$$

$$R_2 - R_1$$

$$R_3 - R_1$$

$$R_4 - R_1$$

Expanding from C_1

$$\begin{vmatrix} 2-x & 2^2-x^2 & 2^3-x^3 \\ 3-x & 3^2-x^2 & 3^3-x^3 \\ 4-x & 4^2-x^2 & 4^3-x^3 \end{vmatrix} = 0$$

$$(2-x)(3-x)(4-x) \begin{vmatrix} 1 & 2+x & 4+2x+x^2 \\ 1 & 3+x & 9+3x+x^2 \\ 1 & 4+x & 16+4x+x^2 \end{vmatrix} = 0$$

taking $(2-x), (3-x),$
 $(4-x)$ common from
 R_1, R_2, R_3 resp.

$$(2-x)(3-x)(4-x) \begin{vmatrix} 1 & 2+x & 4+2x+x^2 \\ 0 & 1 & 5+x \\ 0 & 2 & 12+2x \end{vmatrix} = 0$$

$$R_2 - R_1$$

$$R_3 - R_1$$

Expanding from C_1

$$(2-x)(3-x)(4-x) \begin{vmatrix} 1 & 5+x \\ 2 & 12+2x \end{vmatrix} = 0$$

$$(2-x)(3-x)(4-x)(12+2x-10-2/x) = 0$$

$$(2-x)(3-x)(4-x)(2) = 0$$

$$(2-x)(3-x)(4-x) = 0$$

$$\Rightarrow \boxed{x = 2, 3, 4}$$

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(iv)

$$\begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & x+1 & 3 & 4 & 5 \\ 1 & 2 & x+1 & 4 & 5 \\ 1 & 2 & 3 & x+1 & 5 \\ 1 & 2 & 3 & 4 & x+1 \end{vmatrix} = 0$$

Soln-

Given

$$\begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & x+1 & 3 & 4 & 5 \\ 1 & 2 & x+1 & 4 & 5 \\ 1 & 2 & 3 & x+1 & 5 \\ 1 & 2 & 3 & 4 & x+1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & x-1 & 0 & 0 & 0 \\ 0 & 0 & x-2 & 0 & 0 \\ 0 & 0 & 0 & x-3 & 0 \\ 0 & 0 & 0 & 0 & x-4 \end{vmatrix} = 0$$

$$R_2 - R_1$$

$$R_3 - R_1$$

$$R_4 - R_1$$

$$R_5 - R_1$$

Expanding from C_1

$$\begin{vmatrix} x-1 & 0 & 0 & 0 \\ 0 & x-2 & 0 & 0 \\ 0 & 0 & x-3 & 0 \\ 0 & 0 & 0 & x-4 \end{vmatrix} = 0$$

$$(x-1)(x-2)(x-3)(x-4) = 0 \quad \left(\because \text{det. of a diagonal matrix is equal to the product of diagonal elements.} \right)$$

$$\Rightarrow x = 1, 2, 3, 4$$

(v)

$$\begin{vmatrix} x & a & a & a & a \\ a & x & a & a & a \\ a & a & x & a & a \\ a & a & a & x & a \\ a & a & a & a & x \end{vmatrix} = 0$$

Sol.

Given

$$\begin{vmatrix} x & a & a & a & a \\ a & x & a & a & a \\ a & a & x & a & a \\ a & a & a & x & a \\ a & a & a & a & x \end{vmatrix} = 0$$

$$\begin{vmatrix} x+4a & a & a & a & a \\ x+4a & x & a & a & a \\ x+4a & a & x & a & a \\ x+4a & a & a & x & a \\ x+4a & a & a & a & x \end{vmatrix} = 0$$

$$C_1 + (C_2 + C_3 + C_4 + C_5)$$

$$(x+4a) \begin{vmatrix} 1 & a & a & a & a \\ 1 & x & a & a & a \\ 1 & a & x & a & a \\ 1 & a & a & x & a \\ 1 & a & a & a & x \end{vmatrix} = 0 \quad \text{taking } x+4a \text{ common from } C_1$$

$$(x+4a) \begin{vmatrix} 1 & a & a & a & a \\ 0 & x-a & 0 & 0 & 0 \\ 0 & 0 & x-a & 0 & 0 \\ 0 & 0 & 0 & x-a & 0 \\ 0 & 0 & 0 & 0 & x-a \end{vmatrix} = 0$$

$R_2 - R_1$
 $R_3 - R_1$
 $R_4 - R_1$
 $R_5 - R_1$

Expanding from C_1

$$(x+4a) \begin{vmatrix} x-a & 0 & 0 & 0 \\ 0 & x-a & 0 & 0 \\ 0 & 0 & x-a & 0 \\ 0 & 0 & 0 & x-a \end{vmatrix} = 0$$

$$(x+4a)(x-a)(x-a)(x-a)(x-a) = 0$$

$$\Rightarrow x = -4a, a, a, a, a$$

$$(vi) \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0$$

Sol. Given

$$\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0$$

$$\begin{vmatrix} 4+x & 1 & 1 & 1 \\ 4+x & 1+x & 1 & 1 \\ 4+x & 1 & 1+x & 1 \\ 4+x & 1 & 1 & 1+x \end{vmatrix} = 0$$

$$C_1 + (C_2 + C_3 + C_4)$$

$$(4+x) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0$$

taking $4+x$ Common from C_1

$$(4+x) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & x & 0 & 0 \\ 0 & 0 & x & 0 \\ 0 & 0 & 0 & x \end{vmatrix} = 0$$

$$R_2 - R_1$$

$$R_3 - R_1$$

$$R_4 - R_1$$

Expanding from C_1

$$(4+x) \begin{vmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{vmatrix} = 0$$

$$\Rightarrow (4+x) \cdot x^3 = 0$$

$\Rightarrow$

$$x = -4, 0, 0, 0$$

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Q2 Evaluate each of the following determinants:

$$(1) \begin{vmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \dots & a \end{vmatrix}$$

Sol.

Let $\Delta = \begin{vmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \dots & a \end{vmatrix}$

$= \begin{vmatrix} a+(n-1)d & b & b & \dots & b \\ a+(n-1)d & a & b & \dots & b \\ a+(n-1)d & b & a & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a+(n-1)d & b & b & \dots & b \end{vmatrix}$ By $C_1 + (C_2 + C_3 + \dots + C_n)$

$= [a+(n-1)d] \begin{vmatrix} 1 & b & b & \dots & b \\ 1 & a & b & \dots & b \\ 1 & b & a & \dots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & b & b & \dots & b \end{vmatrix}$ taking $a+(n-1)d$ Common from C_1

$= [a+(n-1)d] \begin{vmatrix} 1 & b & b & \dots & b \\ 0 & a-b & 0 & \dots & 0 \\ 0 & 0 & a-b & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a-b \end{vmatrix}$

$R_2 - R_1$
 $R_3 - R_1$
 $\vdots$
 $R_n - R_1$

Expanding from C_1

$$\Delta = [a + (n-1)d] \begin{vmatrix} a-b & 0 & \dots & 0 \\ 0 & a-b & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & a-b \end{vmatrix}$$

$$(a + (n-1)d)(a-b)^{n-1}$$

(ii)

$$\begin{vmatrix} 1-n & 1 & 1 & \dots & 1 \\ 1 & 1-n & 1 & \dots & 1 \\ 1 & 1 & 1-n & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1-n \end{vmatrix}$$

Sol:-

$$\text{Let } \Delta = \begin{vmatrix} 1-n & 1 & 1 & \dots & 1 \\ 1 & 1-n & 1 & \dots & 1 \\ 1 & 1 & 1-n & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1-n \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 0 & \dots & 0 \\ 1 & 1-n & 1 & \dots & 0 \\ 1 & 1 & 1-n & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1-n \end{vmatrix}$$

$$R_1 + (R_2 + R_3 + \dots + R_n)$$

$$\Delta = 0$$

$$\therefore R_1 = 0$$

(iii)

$$\begin{vmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \\ 3 & 4 & 5 & \dots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n & 1 & 2 & \dots & n-2 & n-1 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \\ 3 & 4 & 5 & \dots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n & 1 & 2 & \dots & n-2 & n-1 \end{vmatrix}$$

$$= \begin{vmatrix} \Sigma n & \Sigma n & \Sigma n & \dots & \Sigma n & \Sigma n \\ 2 & 3 & 4 & \dots & n & 1 \\ 3 & 4 & 5 & \dots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n-2 & n-1 & n & \dots & n-4 & n-3 \\ n-1 & n & 1 & \dots & n-3 & n-2 \\ n & 1 & 2 & \dots & n-2 & n-1 \end{vmatrix}$$

Adding R_2 to R_n
in R_1 & denoting
 $1+2+\dots+n$ by Σn

$$= \Sigma n \begin{vmatrix} 1 & 1 & 1 & \dots & 1 & 1 \\ 2 & 3 & 4 & \dots & n & 1 \\ 3 & 4 & 5 & \dots & 1 & 2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ n-2 & n-1 & n & \dots & n-4 & n-3 \\ n-1 & n & 1 & \dots & n-3 & n-2 \\ n & 1 & 2 & \dots & n-2 & n-1 \end{vmatrix}$$

taking Σn
Common from C_1

$$= \sum_n \begin{vmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 2 & 1 & 2 & \dots & n-2 & -1 \\ 3 & 1 & 2 & \dots & 2 & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ n-2 & 1 & 2 & \dots & -2 & -1 \\ n-1 & 1 & 2-n & \dots & -2 & -1 \\ n & 1-n & 2-n & \dots & -2 & -1 \end{vmatrix}$$

$$C_2 - C_1$$

$$C_3 - C_1$$

$$C_4 - C_1$$

$$C_n - C_1$$

Expanding from R_1

$$= \sum_n \begin{vmatrix} 1 & 2 & \dots & n-2 & n-1 \\ 1 & 2 & \dots & 2 & -1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 2 & \dots & -2 & -1 \\ 1 & 2-n & \dots & -2 & -1 \\ 1-n & 2-n & \dots & -2 & -1 \end{vmatrix}$$

$$= \sum_n \begin{vmatrix} 0 & 0 & \dots & 0 & -1 \\ 0 & 0 & \dots & -n & -1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -n & -1 \\ 0 & -n & \dots & -2 & -1 \\ -n & -n & \dots & -2 & -1 \end{vmatrix}$$

$$C_1 + C_{n-1}$$

$$C_2 + 2C_{n-1}$$

$$C_{n-2} + (n-2)C_{n-1}$$

$$= \sum_n (-1)^{n-1} (n)^{n-2} \begin{vmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 1 \\ 0 & 1 & \dots & 1 & 1 \\ 1 & 1 & \dots & 1 & 1 \end{vmatrix}$$

taking $-n$

Common factor C_{n-2}

$\& -1$ Common
from $C_1, C_2, \dots, C_{n-1}$

Going through each of the preceding columns, ⁷⁴ $(n-1)$ th column shifted to the place of first column, there will be $n-2$ changes of sign. The second last column which now is at the $(n-1)$ th position shifted similarly at the position of second column, there will be $n-3$ changes of sign etc.

So we have

$$\Delta = \sum_{n=1}^{n-1} (-1)^{n-2+1} \cdot (n)$$

1	0	...	0	0
1	1	...	0	0
1	1	...	0	0
1	1	...	1	0
1	1	...	1	1

$$\Delta = \sum_{n=1}^{n-1} (-1)^{n-2+1} \cdot (n) \cdot 1$$

$$\begin{aligned} \text{Now } (n-1)+1 &= (n-1) + (n-2) + (n-3) + \dots + 3 + 2 + 1 \\ &= \frac{(n-1)(n-1+1)}{2} \\ &= \frac{(n-1)(n)}{2} \\ &= \frac{n(n-1)}{2} \end{aligned}$$

So last eq. becomes

$$\Delta = \sum_{n=1}^{n-1} (-1)^{\frac{n(n-1)}{2}} \cdot (n)$$

(iv)

$$\begin{vmatrix} x+1 & x & \dots & x \\ x & x+2 & \dots & x \\ \vdots & \vdots & \ddots & \vdots \\ x & x & \dots & x+n \end{vmatrix}$$

Sol.

Let $\Delta =$

$$\begin{vmatrix} x+1 & x & \dots & x \\ x & x+2 & \dots & x \\ \vdots & \vdots & \ddots & \vdots \\ x & x & \dots & x+n \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \dots & 0 & -n \\ 0 & 2 & \dots & 0 & -n \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & n-1 & -n \\ x & x & \dots & x & x+n \end{vmatrix}$$

$R_1 - R_n$
 $R_2 - R_n$
 $\vdots$
 $R_{n-1} - R_n$

$$= \begin{vmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & n-1 & 0 \\ x & x & \dots & x & nx(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \frac{1}{x}) \end{vmatrix}$$

$C_n + (nC_1 + \frac{n}{2}C_2 + \frac{n}{3}C_3 + \dots + \frac{n}{n-1}C_{n-1})$

$$\Delta = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n-1 \cdot nx(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \frac{1}{x}) = n! \cdot x \cdot (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \frac{1}{x})$$

Q3 If A & B are 3×3 matrices such that
 $\det(A^2 B^3) = 108$ & $\det(A^3 B^2) = 72$.
 Find $\det(2A)$ & $\det(B')$.

Sol. Given

$$\left. \begin{aligned} \det(A^2 B^3) &= 108 \\ \& \det(A^3 B^2) &= 72 \end{aligned} \right\}$$

By product theorem

$$\left. \begin{aligned} \det(A^2) \cdot \det(B^3) &= 108 \\ \det(A^3) \cdot \det(B^2) &= 72 \end{aligned} \right\}$$

or

$$(\det A)^2 \cdot (\det B)^3 = 108 \quad \text{--- (1)}$$

$$\& (\det A)^3 \cdot (\det B)^2 = 72 \quad \text{--- (2)}$$

Dividing (1) by (2)

$$\frac{\det B}{\det A} = \frac{108}{72}$$

$$\text{or } \frac{\det B}{\det A} = \frac{3}{2}$$

$$\Rightarrow \det B = \frac{3}{2} \det A$$

Put in (2)

$$(\det A)^3 \cdot \left(\frac{3}{2} \det A\right)^2 = 72$$

$$\frac{9}{4} (\det A)^5 = 72$$

$$\Rightarrow (\det A)^5 = \frac{72 \times 4}{9}$$

$$(\det A)^5 = 32$$

$$\Rightarrow \boxed{\det A = 2}$$

$$\text{Now } \det(2A) = 2^3 \cdot \det A$$

$$\det(2A) = 8 \times 2$$

So

$$\boxed{\det(2A) = 16}$$

Now

$$\begin{aligned} \det(\bar{B}) &= (\det B)^{-1} \\ &= \left(\frac{3}{2} \det A\right)^{-1} \\ &= \left(\frac{3}{2} \times 2\right)^{-1} \\ &= (3)^{-1} \end{aligned}$$

$$\text{So } \det(\bar{B}) = \frac{1}{3}$$

Q4 Let A be an $n \times n$ matrix. Show that

- (i) $\det A^m = (\det A)^m$ for any +ve integer m
- (ii) If $\det A^m = 1$ then $\det A = \pm 1$
- (iii) If $\det A^m = 0$ then $\det A = 0$

Sol.

(i) We will prove

$$\det A^m = (\det A)^m \text{ by applying induction on } m.$$

Step ① Let $m = 1$

$$\text{So } \det A^1 = (\det A)^1$$

$$\text{or } \det A = \det A$$

Hence it is true for $m = 1$

Step ② Suppose it is true for $m = k$

$$\text{i.e., } \det A^k = (\det A)^k \quad \text{--- ①} \quad \text{for } k \geq 1$$

Step ③ Now we prove it for $m = k+1$

Now

$$\begin{aligned}
 \det(A^{k+1}) &= \det(A^k \cdot A) \\
 &= (\det A^k) \cdot (\det A) \\
 &= (\det A)^k \cdot (\det A)
 \end{aligned}$$

By product theorem

using ①

$$\therefore \det(A^{k+1}) = (\det A)^{k+1}$$

So it is true for $m = k+1$

Hence

$$\det(A^m) = (\det A)^m \quad \text{for all +ve integers } m.$$

(ii) If $\det A^m = 1$ then $\det A = \pm 1$ Soln.

Since $\det A^m = 1$

s.

$$(\det A)^m = 1$$

s.

$$\det A = \pm 1$$

where m is an even integer(iii) If $\det A^m = 0$ then $\det A = 0$ Soln.

Since $\det A^m = 0$

$$\Rightarrow (\det A)^m = 0$$

$$\Rightarrow \det A = 0$$

Q5 For any non-singular matrix C , show that

(i) $\det(\bar{C}) = (\det C)^{-1}$

(ii) $\det(CA\bar{C}) = \det A$

Sol.

(i) Since C is a non singular matrix,
so C^{-1} exists such that

$$CC^{-1} = I$$

$$\Rightarrow \det(CC^{-1}) = \det(I)$$

$$\text{or } \det(C) \cdot \det(C^{-1}) = \det(I) \quad (\text{By product theorem})$$

$$\det C \cdot \det(C^{-1}) = 1$$

$$\det(C^{-1}) = \frac{1}{\det C}$$

$$\det(C^{-1}) = (\det C)^{-1}$$

$$(ii) \quad \det(CAC^{-1}) = \det A$$

Sol.

using product theorem

$$\det(CAC^{-1}) = \det C \cdot \det A \cdot \det C^{-1}$$

$$= \det C \cdot \det C^{-1} \cdot \det A$$

$$= \det(CC^{-1}) \cdot \det A$$

$$= \det I \cdot \det A$$

$$= 1 \cdot \det A$$

$$\text{so } \det(CAC^{-1}) = \det A$$

$\because$ det. of an element
is an element of
field & so they
commute.

By product theorem

Q6 For what value of α is the matrix

$$A = \begin{bmatrix} -\alpha & \alpha-1 & \alpha+1 \\ 1 & 2 & 3 \\ 2-\alpha & \alpha+3 & \alpha+7 \end{bmatrix} \quad \text{singular?}$$

Sol.

Given $A = \begin{bmatrix} -\alpha & \alpha-1 & \alpha+1 \\ 1 & 2 & 3 \\ 2-\alpha & \alpha+3 & \alpha+7 \end{bmatrix}$

Since A is singular

$$\text{So } \det A = 0$$

$$\Rightarrow \begin{vmatrix} -\alpha & \alpha-1 & \alpha+1 \\ 1 & 2 & 3 \\ 2-\alpha & \alpha+3 & \alpha+7 \end{vmatrix} = 0$$

$$\begin{vmatrix} -\alpha & 3\alpha-1 & 4\alpha+1 \\ 1 & 0 & 0 \\ 2-\alpha & 3\alpha-1 & 4\alpha+1 \end{vmatrix} \quad \begin{array}{l} C_2 - 2C_1 \\ C_3 - 3C_1 \end{array}$$

Expanding from R_2

$$- \begin{vmatrix} 3\alpha-1 & 4\alpha+1 \\ 3\alpha-1 & 4\alpha+1 \end{vmatrix} = 0$$

$$-(3\alpha-1)(4\alpha+1) \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

obviously matrix A is singular for all values of α .

$$\frac{\text{Adj } A}{\det A} \cdot A = \frac{-1}{28} \begin{bmatrix} -2 & -2 & -9 \\ 20 & -8 & 6 \\ -2 & -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 4 & 1 & 6 \\ 2 & 0 & -2 \end{bmatrix}$$

$$= \frac{-1}{28} \begin{bmatrix} -2-8-18 & 2-2+0 & -6-12+18 \\ 20-32+12 & -20-8 & 60-48-12 \\ -2-8+10 & 2-2+0 & -6-12-10 \end{bmatrix}$$

$$= \frac{-1}{28} \begin{bmatrix} -28 & 0 & 0 \\ 0 & -28 & 0 \\ 0 & 0 & -28 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{\text{Adj } A}{\det A} \cdot A = I \quad \text{②}$$

from ① & ②

$$A \cdot \frac{\text{Adj } A}{\det A} = \frac{\text{Adj } A}{\det A} \cdot A = I$$

$\Rightarrow$

$$A^{-1} = \frac{\text{Adj } A}{\det A}$$

Q8 Evaluate

$$\begin{vmatrix} 1 & 0 & -1 & 2 \\ 2 & 3 & 2 & -2 \\ 2 & 4 & 2 & 1 \\ 3 & 1 & 5 & -3 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 1 & 0 & -1 & 2 \\ 2 & 3 & 2 & -2 \\ 2 & 4 & 2 & 1 \\ 3 & 1 & 5 & -3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 & 0 \\ 2 & 3 & 4 & -6 \\ 2 & 4 & 4 & -3 \\ 3 & 1 & 8 & -9 \end{vmatrix}$$

$C_3 + C_1$

$C_4 - 2C_1$

Expanding from R_1

$$= \begin{vmatrix} 3 & 4 & -6 \\ 4 & 4 & -3 \\ 1 & 8 & -9 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -20 & 21 \\ 0 & -28 & 33 \\ 1 & 8 & -9 \end{vmatrix}$$

$R_1 - 3R_3$

$R_2 - 4R_3$

Expanding from C_1

$$= \begin{vmatrix} -20 & 21 \\ -28 & 33 \end{vmatrix}$$

$$= -660 + 588$$

$$\Delta = -72$$